TEST - verification

DB-HALL

Customer	TEST
Job number	TEST-001
Issued under	iec-default
Design rules	software defaults, not confirmed by the company's engineers

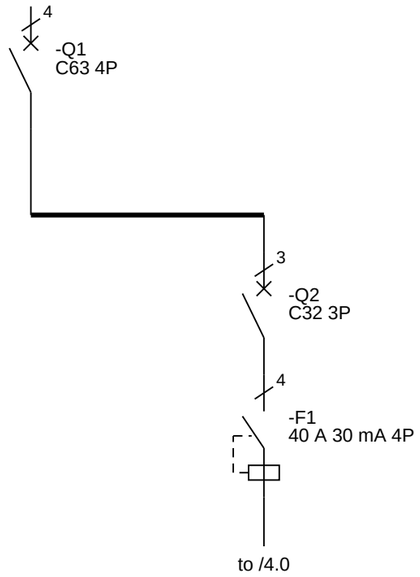
Rev.	Date	Description	Drawn	Checked	Approved
A	2026-10-23	TEST issue	V4		

Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001		**Title page**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			1 of 9	

Page	Title	Board
1	Title page	
3	Main power	DB-HALL
4	Distribution loads	DB-HALL
5	Design notes	DB-HALL
6	Layout	DB-HALL
7	Terminals	DB-HALL
8	Cable list	
9	Parts list	

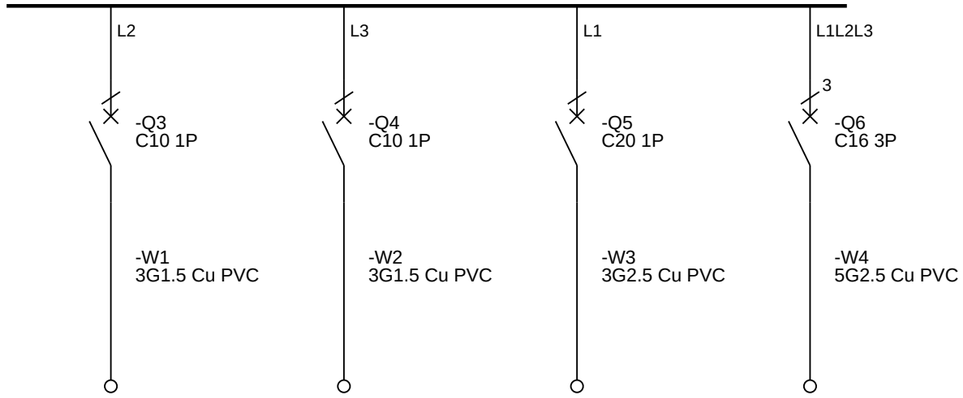
Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001		**Table of contents**	A
Date	Drawn	Checked	Approved	Page	
2026-10-23	V4			2 of 9	

3/N/PE ~ 50 Hz
400 V TN-S



Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001	DB-HALL	**Main power**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			3 of 9	

from -F1 40 A 30 mA 4P /3.1

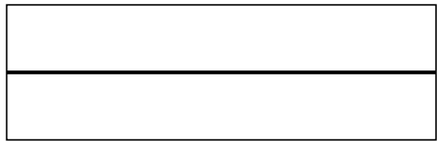


Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001	DB-HALL	**Distribution loads**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			4 of 9	

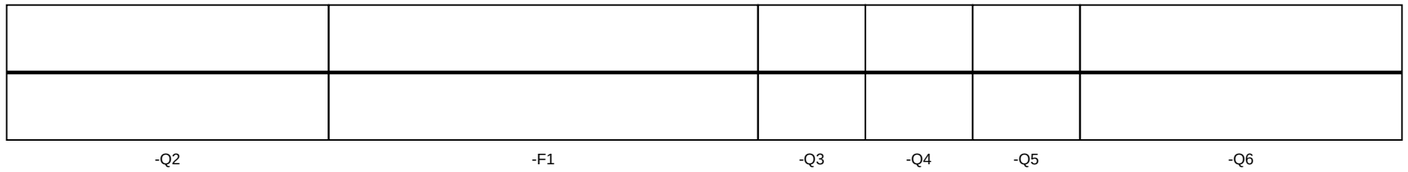
No.	Note
1	Sockets hall east (TEST VALUE): Ib 16.84 A exceeds the company's 16 A for socket; rated 20 A instead.
2	Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination.
3	Phase imbalance 44.5 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
4	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
5	No fault level given: every breaker's breaking capacity is left to be confirmed.
6	Leave 1 spare outgoing ways (20 %).
7	Loads without a power factor were taken at cos phi 0.9.
8	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
9	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
10	Circuit rules are this software's defaults, not confirmed by the company's engineers.
11	Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 15 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001	DB-HALL	**Design notes**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			5 of 9	

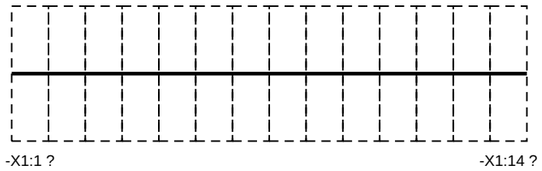
Incomer: 70 mm of rail



Residual current group: 227.5 mm of rail



Terminals: 0 mm of rail; 14 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001	DB-HALL	**Layout**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			6 of 9	

Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L2	Hall lighting row 1 (TEST VALUE)	-W1	1.5	8WH1000-0AF00
-X1:2	N	Hall lighting row 1 (TEST VALUE)	-W1	1.5	8WH1000-0AF00
-X1:3	PE	Hall lighting row 1 (TEST VALUE)	-W1	1.5	8WH1000-0CF07
-X1:4	L3	Hall lighting row 2 (TEST VALUE)	-W2	1.5	8WH1000-0AF00
-X1:5	N	Hall lighting row 2 (TEST VALUE)	-W2	1.5	8WH1000-0AF00
-X1:6	PE	Hall lighting row 2 (TEST VALUE)	-W2	1.5	8WH1000-0CF07
-X1:7	L1	Sockets hall east (TEST VALUE)	-W3	2.5	8WH1000-0AF00
-X1:8	N	Sockets hall east (TEST VALUE)	-W3	2.5	8WH1000-0AF00
-X1:9	PE	Sockets hall east (TEST VALUE)	-W3	2.5	8WH1000-0CF07
-X1:10	L1	AC unit (TEST VALUE)	-W4	2.5	8WH1000-0AF00
-X1:11	L2	AC unit (TEST VALUE)	-W4	2.5	8WH1000-0AF00
-X1:12	L3	AC unit (TEST VALUE)	-W4	2.5	8WH1000-0AF00
-X1:13	N	AC unit (TEST VALUE)	-W4	2.5	8WH1000-0AF00
-X1:14	PE	AC unit (TEST VALUE)	-W4	2.5	8WH1000-0CF07

Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001	DB-HALL	**Terminals**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			7 of 9	

Cable	From	To	Type	Length
DB-HALL -W1	DB-HALL -Q3	Hall lighting row 1 (TEST VALUE)	3G1.5 Cu PVC	30 m
DB-HALL -W2	DB-HALL -Q4	Hall lighting row 2 (TEST VALUE)	3G1.5 Cu PVC	35 m
DB-HALL -W3	DB-HALL -Q5	Sockets hall east (TEST VALUE)	3G2.5 Cu PVC	25 m
DB-HALL -W4	DB-HALL -Q6	AC unit (TEST VALUE)	5G2.5 Cu PVC	20 m

Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001		**Cable list**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			8 of 9	

Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 4P	not selected		-Q1
1	Circuit-breaker C32 3P	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 4P	not selected		-F1
2	Circuit-breaker C10 1P	not selected		-Q3, -Q4
1	Circuit-breaker C20 1P	not selected		-Q5
1	Circuit-breaker C16 3P	not selected		-Q6
10	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	DB-HALL -X1
4	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	DB-HALL -X1

Company	Project	Job number	Board	Title of page	Rev.
	TEST - verification	TEST-001		**Parts list**	A
Date	**Drawn**	**Checked**	**Approved**	**Page**	
2026-10-23	V4			9 of 9	